MILITARY SPECIFICATION

ZINC CARBONATE, ANALYZED REAGENT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one analyzed reagent grade of zinc carbonate.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel Strapping, Flat.
TT-W-572	-	Wood-Preservative; Water-Repellent.
PPP-B-585	-	Boxes, Wood, Wirebound.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-C-186	-	Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.
PPP-C-301	-	Chemicals, Dry and Paste, Packaging and Packing of.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

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MIL-P-15011	-	Pallets, Material Handling, Wood, Post Construction, 4-Way Entry.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" 4-Way (Partial) Pallet Skids, Runners, or Pallet-Type Base.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Chemical characteristics. Zinc carbonate shall conform to the chemical characteristics of table I when tested as specified therein.

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Table I. Chemical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay (as ZnO)	70.0	----	4.2.4.1
Insolubles in sulfuric acid	----	0.020	4.2.4.2
Chloride (Cl)	----	0.002	4.2.4.3
Nitrate (NO_3)	----	0.005	4.2.4.4
Sulfate (SO_4)	----	0.010	4.2.4.5
Heavy metals (as Pb)	----	0.005	4.2.4.6
Iron (Fe)	----	0.005	4.2.4.7
Substances not precipitated by ammonium sulfide	----	0.50	4.2.4.8

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services comform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the zinc carbonate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

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4.2.2.2 For test. Sampling for test shall be conducted in accordance with table II. A representative specimen of approximately 1/4 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

Table II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2-25	2
26-150	3
151-1,200	5
1,201-7,000	8
7,001-20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard packing aids not as specified
(f) Strapping or tape banding not as specified
(g) Wood preservation incorrect or missing (level A packing only)
(h) Marking incorrect, missing, or illegible

4.2.3.2 For test. Approximately equal portions of all of the specimens taken in 4.2.2.2 shall be thoroughly mixed to form a composited specimen of no less than 1/4 pound. The composited specimen shall be tested as specified in 4.2.4. Each test shall be conducted in duplicate analysis. Failure of either analysis of any test shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Assay (as Zno). Weigh to the nearest 0.1 milligram (mg) approximately 2 grams (g) of the specimen and dissolve in 50 milliliters (ml) of 1N sulfuric acid. Titrate the excess acid with 1N sodium hydroxide solution using methyl orange indicator. Calculate the percent zinc oxide as follows:

$$\text{Percent zinc oxide} = \frac{4.069\ (AB - CD)}{W}$$

where: A = Milliliters of sulfuric acid added,
B = Normality of sulfuric acid,
C = Milliliters of sodium hydroxide solution used,
D = Normality of sodium hydroxide solution, and
W = Weight of specimen in grams.

4.2.4.2 Insolubles in sulfuric acid. Weigh to the nearest milligram approximately 10 g of the specimen into a beaker. Add, in small portions, a mixture of 100 ml of water and 7 ml of concentrated sulfuric acid and heat on a steam bath for 1 hour. Filter through a filter crucible weighed to the nearest 0.1 mg, wash with hot water, dry to constant weight at 105° ± 2° C, cool to room temperature in a desiccator, and weigh to the nearest 0.1 mg. Calculate the percent insolubles in sulfuric acid as follows:

$$\text{Percent insolubles in sulfuric acid} = \frac{100(A - B)}{W}$$

where: A = Weight of crucible and residue in grams,
B = Weight of crucible in grams, and
W = Weight of specimen in grams.

4.2.4.3 Chloride. Weigh 1.000 g of the specimen and dissolve in a mixture of 2 ml of concentrated nitric acid and 20 ml of water. Add 1 ml of 1.7-percent silver nitrate solution. Any turbidity produced shall be no greater than that produced when 0.02 mg of chloride ion is tested in the same manner as the specimen.

4.2.4.4 Nitrate. Dissolve 0.33 g of the specimen in 10 ml of water. Add 20 mg of sodium chloride and 0.2 ml of diphenylamine solution (prepared by dissolving 100 mg of diphenylamine in 25 ml of concentrated sulfuric acid and 25 ml of water, adding 2 ml of concentrated hydrochloric acid, and adding concentrated sulfuric acid to make 100 ml). Slowly add 10 ml of concentrated sulfuric acid. Mix gently and allow to stand for 30 minutes. No blue color shall be present.

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4.2.4.5 Sulfate. Weigh 1.000 g of the specimen and dissolve in 5 ml of water and the minimum amount of concentrated hydrochloric acid necessary. Evaporate the solution to 2 ml, take up with 1 ml of 1N hydrochloric acid and 20 ml of water, and add 2 ml of 10-percent barium chloride solution. Any turbidity produced shall be no greater than that produced in a control containing 0.1 mg of sulfate ion, 1 ml of 1N concentrated hydrochloric acid, and 2 ml of 10-percent barium chloride solution in the same final volume as the specimen.

4.2.4.6 Heavy metals (as Pb). Weigh 1.000 g of the specimen and heat with 10 ml of water. Add concentrated nitric acid dropwise until the specimen is just dissolved. Add concentrated ammonium hydroxide solution dropwise until a slight permanent precipitate is produced, then add just sufficient concentrated nitric acid to dissolve the precipitate and dilute to 20 ml with water. To 5 ml of the solution, add 0.025 mg of lead ion and 30 ml of 10-percent sodium cyanide solution and dilute to 50 ml with water. (This is solution A.) To the remaining 15 ml of specimen solution, add 30 ml of 10-percent sodium cyanide solution. (This is solution B.) Add 0.5 ml of freshly prepared 5-percent sodium sulfide solution to each solution. Any color produced in solution B shall be no darker than that produced in solution A.

4.2.4.7 Iron. Weigh 1.000 g of the specimen and heat with 5 ml of water. Add concentrated hydrochloric acid dropwise until just dissolved. Cool to room temperature and dilute to 50 ml with water. Add 2 ml of concentrated hydrochloric acid, 30 mg of ammonium persulfate, and 3 ml of 25-percent ammonium thiocyanate solution. Any red color produced shall be no darker than that produced in a control containing 0.05 mg of iron ion, 2 ml of concentrated hydrochloric acid and the same quantities of the other reagents used in the specimen test in the same final volume as the specimen.

4.2.4.8 Substances not precipitated by ammonium sulfide. Weigh to the nearest 0.1 mg approximately 1 g of the specimen, dissolve in 10 ml of water and 2 ml of concentrated sulfuric acid, and dilute to 90 ml with water. Add 10 ml of concentrated ammonium hydroxide solution and precipitate the zinc with hydrogen sulfide. Filter without washing. Evaporate the filtrate to dryness and ignite to constant weight. Cool to room temperature in a desiccator and weigh to the nearest 0.1 mg. Calculate the percent substances not precipitated by ammonium sulfide as follows:

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Percent substances not precipitated by ammonium sulfide = $\frac{100\ A}{W}$

where: A = Weight of filtrate residue in grams, and
W = Weight of specimen in grams.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound + 0.1 ounce of zinc carbonate shall be packaged in accordance with the applicable requirements of PPP-C-301, in a type I, class 1, wide-mouth glass bottle, in inclosure b as specified therein. The bottle shall also conform to glass type b, light penetration grade optional of PPP-C-186. Closure of the bottle shall be with a plastic screw cap and a secondary seal of tape or A band conforming to PPP-C-186.

5.1.1.1 Intermediate packaging. Twelve bottles of zinc carbonate, packaged as specified in 5.1.1, shall be intermediately packaged as specified in PPP-C-301 in a snug-fitting fiberboard box except that the bottles shall be arranged in three rows of four bottles each.

5.1.2 Level C. One pound + 0.1 ounce of zinc carbonate shall be packaged as specified in 5.1.1 except that the bottle shall not be placed in inclosure b of PPP-C-301.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Four intermediate packages of zinc carbonate, prepared as specified in 5.1.1.1, shall be packed upright in a wood box conforming to either type overseas, style I or J of PPP-B-601; or type 3, style optional of PPP-B-585. Fiberboard conforming to grade W5c of PPP-F-320 shall be used for pads and fillers as specified in PPP-C-301 to assure a tight pack. Each box shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification. Prior to filling and marking, each box or the unfinished wood parts thereof shall be completely immersed for a minimum of 3 minutes in a water-repellent wood preservative conforming to composition A of TT-W-572.

5.2.2 Level B. Zinc carbonate shall be packed as specified in 5.2.1 except that the box shall conform to style RSC, grade V3c of PPP-B-636. The box shall be closed in accordance with the requirements for taping style RSC boxes as exterior containers and subsequently reinforced with tape banding as specified in the appendix to PPP-B-636.

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5.2.3 Level C. Twelve bottles of zinc carbonate, packaged as specified in 5.1.2, shall be packed in a fiberboard box conforming to style RSC, grade W5c of PPP-B-636. The box shall be fitted with half-slotted fiberboard separators which shall form snug-fitting cells for each of the bottles and which shall be the same height as the bottles. All inner faces of the box shall be lined with full box-size fiberboard pads. Fiberboard used for separators and pads shall conform to grade W5c of PPP-F-320. The box shall be closed in accordance with the requirements for taping specified in the appendix to PPP-B-636.

5.3 Marking. In addition to the marking for unit containers specified in 5.3.1 any marking required by the contract or order (see 6.2), unit and intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Special marking. Each unit container shall be durably and legibly labeled to show the following special information:

(a) Empirical formula.
(b) Formula weight.
(c) Manufacturer's lot number.
(d) Assay.
(e) Analysis or maximum limits of impurities.
(f) Net weight of contents.
(g) Precautionary marking:

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5.4 Palletization. When specified in the contract or order (see 6.2), shipping containers shall be palletized in accordance with the applicable load type specified in MIL-STD-147 using pallets conforming to type I, style 1 of MIL-P-15011. Tie-down strapping, 1.25- by 0.035-inch, shall conform to type I, class B, grade 1 of QQ-S-781. One unitizing strap, 0.75- by 0.023-inch, conforming to type I, class B, grade 1 of QQ-S-781, shall be used around each tier of boxes. The pallet or all unfinished wood parts thereof shall be treated with preservative as specified for the wood boxes in 5.2.1.

6. NOTES

6.1 Intended use. Zinc carbonate is intended for use as a laboratory reagent.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required (see 5.1 and 5.2).
(c) Additional marking, if required (see 5.3).
(d) Palletization, if required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

Custodians:

Army - MU
Air Force - 68

Review activities:

Army - MD, MU(FA)
Air Force - 68
DSA - GS

User activities:

Army - MI, SM

Preparing activity:

Army - MU(EA)

Project No. 6810-0460

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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OFFICIAL BUSINESS

Commanding Officer
Edgewood Arsenal
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Edgewood Arsenal, Maryland 21010

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